

ig: $K > 0$ & $\lim_{t \rightarrow \infty} p_{b,t,\infty}(K,t) = p_{b,\infty}(K)$

$$p_{b,\infty}(K) = \tilde{z} [\lambda_b (\lambda_b (K-1) + \mu_b) p_{b,\infty}(K-1) - (\lambda_b K + \mu_b) p_{b,\infty}(K)]$$

$$\Rightarrow p_{b,\infty}(K) [\tilde{z} (\lambda_b K + \mu_b) + 1] = p_{b,\infty}(K-1) [\tilde{z} (\lambda_b K + (\mu_b - \lambda_b))]$$

$$\Rightarrow \frac{p_{b,\infty}(K)}{p_{b,\infty}(K-1)} = \frac{\tilde{z} \lambda_b K + \tilde{z} \mu_b - \tilde{z} \lambda_b}{\tilde{z} \lambda_b K + \tilde{z} \mu_b + 1} = \frac{K + \frac{\mu_b}{\lambda_b} - 1}{K + \frac{\mu_b}{\lambda_b} + (\frac{\lambda_b}{\mu_b})}$$

$$\Rightarrow p_{b,\infty}(K) = \frac{A_b \Gamma(K + \frac{\mu_b}{\lambda_b})}{\Gamma(K + \frac{\mu_b}{\lambda_b} + (1 + \frac{\lambda_b}{\mu_b}))} \quad (6)$$

$K \geq 0$ & $\lim_{t \rightarrow \infty}$:

$$p_{b,\infty}(K=0) = \tilde{z} [\lambda_b K + \mu_b + 1] \frac{0}{p_{b,\infty}(-1)} - (\lambda_b K + \mu_b) p_{b,\infty}(K=0) + m(1-g) \delta_{0,0}$$

$$\Rightarrow p_{b,\infty}(0) [\tilde{z} \mu_b + 1] = m(1-g) \Rightarrow p_{b,\infty}(0) = \frac{m(1-g)}{\tilde{z} \mu_b + 1}$$

ig (6) also $\approx$ holds at $K=0$ (*, why would it?)

$$\Rightarrow A_b = \frac{m(1-g) \Gamma(\frac{\mu_b}{\lambda_b} + (1 + \frac{\lambda_b}{\mu_b}))}{(1 + \mu_b \tilde{z}) \Gamma(\frac{\mu_b}{\lambda_b})}$$

& conjecture by symmetry:

$$A_a = \frac{m \tilde{z} \Gamma(\frac{\mu_a}{\lambda_a} + (1 + \frac{\lambda_a}{\mu_a}))}{(1 + \mu_a \tilde{z}) \Gamma(\frac{\mu_a}{\lambda_a})} //$$